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Contents, Abstracts and Keywords

The Political Logic of the "China Miracle": An Ecological Institutionalist Study on the Chinese Path to Modernization …… **Zhao Kejin**(3)

Abstract: Since the reform and opening up, the Chinese path to modernization has created two major miracles: rapid economic development and long-term social stability, thus resolving the coexistence dilemma of stability and development in western political science. The political logic behind the "China Miracle" has also triggered intense debate in academia. From the perspective of Ecological Institutionalism, the reason why Chinese path to modernization has achieved these two miracles is mainly because of the combination of socialism with Chinese characteristics and the ecological conditions specific to its time and space. In terms of institutional attributes, China has established a new system that unifies the leadership of the CPC, the running of the country by the people, and law-based governance, advancing the modernization of China's system and capacity for governance, thereby providing a powerful stabilizer for Chinese society. In terms of the ecological system, China has integrated Marxism with its specific realities and fine traditional culture, aligning China's development with global common development to promote the construction of an open economic system and an open world economy, thereby unleashing tremendous developmental momentum for the Chinese economy. As China embarks on a new journey toward fully building a modern socialist country, it should persevere in coordinating domestic and external situations, balancing development and security, and constantly improve governance of a major country as well as a major party while participating in and even leading the economic globalization. By doing so, China is supposed to promote the great rejuvenation of the Chinese nation through the Chinese path to modernization, continuously creating the "China Miracle" of rapid economic development and long-term social stability. This is the core political logic of the Chinese path to modernization.

Key words: China Miracle; ecological institutionalism; institutional attributes; ecological system; Chinese path to modernization

Reproduction of Order: Based on the Party's Experience in Supervising Officials during the Chinese Modernization Process …… **Wang Dongqi**(20)

Abstract: The reproduction of order is a core issue of modernization and also a critical dimension of political transformation. Since 1840, confronted with the disintegration of the traditional order, Chinese society has grappled with a comprehensive confidence crisis spanning material technology, systems, and culture. In the quest for national salvation and rejuvenation, modern China underwent a profound social revolution, ultimately forging a distinctive path to modernization that achieves national independence, prosperity, and popular wellbeing. This study examines the Communist Party of China (CPC) supervision of officials as a central mechanism within Chinese modernization process, exploring its role in addressing the fundamental challenge of reproducing order. Party supervision of officials provides the primary force essential for order reproduction. Through this mechanism, the CPC has effectively initiated and continually deepened three concrete processes amidst comprehensive transformations across the value system, social system, and power structure: institutional power creation, meritocratic selection and appointment, institutional integration. This has yielded a viable modernization pathway grounded in the realities of China that successfully enables order reproduction. Building upon China's experience, this research further analyzes the theoretical significance of the Party supervision of officials in understanding order reproduction. It utilizes this distinctively Chinese iconic concept to transcend Western-centric theoretical dichotomies-such as "tradition vs. modernity", "West vs. China", and "civilized world vs. non-civilized world". Moreover, it highlights the critical importance for developing nations in their modernization journeys of constructing autonomous principal force and identifying endogenous coupling mechanisms that integrate their unique historical contexts and humanistic traditions within their value systems, social systems, and power structures.

Key words: order; Party supervision of officials; civilization; governance system

China's Consensus-oriented Governance: A Typological Analysis of State Governance based on Political Function …… **Xie Zhikui**(33)

Abstract: "Striving in Unity" is one of the distinctive characteristics of CPC's governance, thus on which is of great theoretical and realistic significance to make an academical analysis. Unlike Aristotle's typology on regimes, which is mainly based on political structure features, this research tries to reveal the functional side. From two dimensions and four quadrants of whether or not having fully

political expression and integrated political outputs, this research classifies different regimes into four ideal types: Jungle Governance, Despotic Governance, Boycotting Governance, and Consensus-oriented Governance. This research argues that capitalist party politics operates on the basis of distributional coalitions chasing their respective interests, hence is easy to stuck into Boycotting Governance featured with "oppose for opposing" that caused governance failure. The Striving in Unity governance pattern of the CPC, however, reflects the distinctive feature and institutional advantages of Consensus-oriented Governance. During the whole process of Chinese-style modernization, we must emphasize consensus-oriented governance and realize the dialectical unity of opinion expression, interest integration and political output.

Key words: striving in unity; consensus-oriented governance; boycotting governance; state governance pattern; governance performance

The Challenge of government-interdepartmental collaboration and Its Reflection

…… **Kong Fanbin, Xu Fei**(46)

Abstract: In the process of accelerating the modernization of social governance, government-interdepartmental collaboration is a common way for the government to fulfill its responsibilities. Although government-interdepartmental collaboration has become the general trend of government governance, the specialization of government departments has not weakened, but shows the overlapping characteristics of specialization and government-interdepartmental collaboration. The key to incentivizing inter-departmental cooperation depends on the reorganization of power and the reconstruction of legitimacy following "trans-boundary organization" activities. The premise of its success is to break through the vertical control path of the bureaucracy described in the classical organization theory and turn to the establishment of a horizontal trans-organizational boundary incentive model. Although refining the effective incentive mechanism of cross-departmental collaboration from case studies is helpful to practice development, the deep-seated problems of cross-departmental collaboration incentive in government cannot be solved purely through empirical research. An ethical enlightenment is imperative to transform the vertical control inertia of government and to recognize that horizontal collaboration incentive has become a function of overall organizational performance.

Key words: incentive; government-interdepartmental collaboration; bureaucracy; collective action; knowledge production

The Enlarged Session: Institutional Functions in the CPC's Policy-Making Process

…… **Chen Jiagang, Chen Lingyu**(61)

Abstract: Democratic centralism constitutes the fundamental organizational and leadership principle of the Communist Party of China (CPC). "Conference-based decision-making" serves as a vital approach through which the CPC implements this principle. Functioning as a highly adaptable decision-making method, the enlarged session has enabled the CPC to overcome structural constraints of regular conferences throughout its century-long struggle. By convening enlarged sessions, the Party has been able to expand the boundaries of the established decision-making frameworks, concentrate on critical issues, and achieve effective decision-making. As a context-specific institutional arrangement, the enlarged session primarily fulfills three core functions: establishing policy legitimacy, resolving policy crises, and articulating the Party's strategic interests. Amidst the ongoing standardization, institutionalization, and proceduralization of the Party and state governance, regular conferences are positioned to assume greater prominence. To optimize decision-making, the CPC must prioritize stable, institutionalized pathways that align with democratic, law-based, and scientific principles.

Key words: the Communist Party of China; enlarged session; decision-making mechanism; democratic centralism; policy process

Evaluative Democracy: Promoting Government Responsiveness via Intergovernmental Competition

…… **Zeng Yu**(74)

Abstract: The Western academia has long regarded competitive elections as the institutional foundation of democracy, generally considering the improvement of electoral systems as the primary goal of democratic development. In reality, elections are not the only path to achieving democracy, nor is electoral competition the sole form of political contestation. Given that government responsiveness is a key attribute of modern democracy and that competition is an important mechanism for realizing such responsiveness, this paper creatively introduces the concept of "evaluative democracy". Through a comparison with electoral democracy, it outlines the inherent logic and political significance of evaluative democracy from a theoretical perspective. It systematically discusses the operational mechanisms, characteristics, and conditions necessary for the functioning of evaluative democracy. On this theoretical basis, the paper conducts an in-depth study of the government hotline reforms implemented in various regions across China, analyzing their democratic attributes, response mechanisms, practical challenges,

and optimization paths, and then offers empirical references and methodological insights on enhancing government responsiveness via intergovernmental competition and on improving the whole-process people's democracy system through evaluative democracy from a practical perspective. Finally, this paper presents a series of recommendations on how to advance evaluative democracy.

Key words: evaluative democracy; intergovernmental competition; government responsiveness; government hotline; whole-process people's democracy

Reshaping and Improving of Grassroots Governance in the Context of State Resource Allocation to Rural Areas …… **He Xuefeng, Gui Hua**(92)

Abstract: In the process of advancing rural vitalization, the state has consistently invested rural support resources to enhance infrastructure and public services. While this influx has markedly improved the rural landscape, it has also reshaped the underlying logic of grassroots governance. The focus of governance has increasingly shifted toward the execution of higher-level administrative tasks, thereby rendering the previous model of village self-governance, which is centered on internal mobilization, progressively unsustainable. In certain regions, formalism has emerged as an issue in grassroots governance. Thus, optimizing the approach to resource allocation and converting state inputs into more effective grassroots governance mechanisms is crucial for rural vitalization and the modernization of rural governance. It is essential that resource allocation to rural areas does not result in the government monopolizing rural affairs. A vibrant and well-organized village is a prerequisite for effective grassroots governance and a foundation for national governance modernization. By establishing a more nuanced resource allocation framework—directing a substantial portion towards mobilizing local communities—grassroots governance will not only be strengthened, but it will also respond to the New Era's call to "stimulate the enthusiasm, initiative, and creativity of the people".

Key words: grassroots governance; state resource allocation to rural areas; formalism; mobilizing local communities; effective governance

Government Labor Costs and County Optimal Population Size: Findings Based on Interpretable Machine Learning Methods in Northeast China

…… **Zhao Jujun, Wang Kun**(106)

Abstract: Due to the combined effects of responsibility isomorphism, inflexible staffing structure,

and the difficulty in realizing economies of scale, less populated counties have become particularly vulnerable to the "subsistence finance" problem and the labor costs are extremely heavy. While promoting consolidation of less populated counties has emerged as one of the possible options, the optimal county population size has not yet been determined through scientific empirical research. Considering regional differences, this paper takes the county-level regions in northeastern China, characterized by significant outward migration and weak financial foundations, as its research subjects. It collects multidimensional data and utilizes the random forest model and SHAP interpretation method in the machine learning approach to explicitly define the optimal population size based on the verification of the key impact of population size on labor costs of county-level government, combined with the ICE plots. The study finds that population size is the primary factor affecting government labor costs and that its impact follows an L-shaped curve: below a population of 450,000, an increase in population size can reduce labor costs per capita, but beyond this threshold, further increases in population size no longer reduce these costs. The main contribution of the paper is to provide a reference for decision-making and methodological guidance for similar research involving forecasting by clarifying the complex relationship between county population size and government labor costs.

Key words: less populated county; population size; administrative division; labor costs; machine learning

Multi-level composite "Co-governance": How Platforms Empower Urban Social Governance ——A "Level-Unit" Analytical Perspective …… **Li Zhonghan, Gao Weixing**(121)

Abstract: Empowering urban governance through platforms is a key component of advancing the modernization of national governance, and research into the corresponding empowerment mechanisms has become a new area of academic growth. Based on the "level-unit" perspective, this paper analyzes the "structure-process" of platform organizations to explore the internal mechanisms of platform empowerment in urban social governance. The research finds that the platform integrates multi-level governments and organizations of "Cities—Single Districts (counties, cities)—Streets (townships)—Communities" into a unified management framework. Based on a nested structure and standardized "input-output" processes, it presents a multi-level composite "co-governance" mechanism that connects all levels and links internal and external entities. As the core mechanism of platform-enabled urban social governance, multi-level composite "co-governance" breaks through the limitation of

vertical hierarchical governance mechanisms and horizontal networked governance mechanisms, which are currently only connected and interacted at the grassroots governance level. It extends these mechanisms upward to higher government levels such as districts and cities, thereby alleviating the tension between the vertical and horizontal mechanisms to a certain extent. This not only breaks through the current challenges of urban social governance but also creates a new urban governance model. However, platform empowerment has certain limitations and risks. Continuously improving the multi-level composite "co-governance" mechanism from both technical and institutional perspectives is the fundamental challenge facing platform empowerment.

Key words: multi-level composite "co-governance"; platform; city-wide social governance; "Level-Unit"; modernization of national governance

"Tianxia-State": An Explanation of the Status of China in the World

…… **Wang Shikai, Chen Yanchao**(135)

Abstract: China is a builder of global peace, a contributor to global development, and a defender of the international order. The fundamental issue underlying this position is the status of China in the world. From the perspective of the integration of Chinese political tradition and human civilization, China's status in the world can be defined as "tianxia-state". With an advanced institutional system and great civilizational achievements as its core content, China possesses many qualities that contribute to the common development of other countries. At the same time, it offers other countries a development prospect different from capitalism. As a continuously innovating civilization, the "tianxia-state" evolves towards a more reasonable form through its development. This means that China's modern civilization embodies the trend of human civilization development and contains the ideal that the human political order will move towards universal harmony. China has become conscious of being a "tianxia-state" in the historical process of great rejuvenation and is simultaneously engaged in external practices centered on promoting the construction of the human community with shared future. China, as a "tianxia-state" within the modern world system, plays a crucial role and holds significant practical implications and theoretical value.

Key words: "Tianxia-State"; China; the status of China in the world; human community with a shared future

National Language of Modern State-Building: A Case Study of the Language Evolution in France …… **Guo Taihui, Xiang Zhiyang**(152)

Abstract: The national language, differing from both world languages and regional dialects, serves as both a vital tool for communication and integration within a nation-state and a symbolic marker of national unity. Consequently, the construction of a national language is both a component and a result of modern state-building. This article uses the evolution of the French language as a case study to illustrate how, after the late Middle Ages, a world language (Latin) gradually declined. French, initially a dialect of the monarchy, rose to become the standard language within the national territory. During the French Revolution, French was promoted as a national language for all citizens, aligning with the ideals of the French Republic and Nation. This process laid the groundwork for a hierarchical order of world language, national language, and regional dialects, pioneering the modern transformation of language. This historical case demonstrates that the construction of a national language is a critical dimension of modern state-building, playing a significant role in national integration and unity. It was facilitated by the dual empowerment of both the state and the nation. A close political mechanism emerged among the state, nation, and language, which, theoretically, offers fertile ground for the development of a new field of language politics, and practically, provides a strategy to counterbalance the dual challenges posed by world languages and regional dialects.

Key words: modern state-building; national language construction; language evolution in French; politics of language

Political Values in the Digital Intelligence Era: Hierarchical Characteristics and Interactive Logic in China …… **Zang Leizhen, Li Wei**(166)

Abstract: Political values constitute individuals' normative orientations toward political life, shaping both political attitudes and behaviors, and serving as a fundamental component for understanding the maintenance of socio-political order. In the era of digital intelligence—characterized by the pervasive application of AI, big data, and algorithmic governance—citizens' political cognition in China is undergoing significant transformation. This study conceptualizes political values as a multi-layered structure, distinguishing between collective political values and individual political values. It explores how these levels interact within technology-embedded processes of political socialization. Findings suggest that Chinese political values are marked by a distinctive equilibrium of "continuity

amid change." On the one hand, digital technologies reinforce collective political values by strengthening national identity; on the other, they foster pluralistic expressions at the individual level. Yet, the accelerating pace of socio-economic development introduces frictions between the inertia of political value formation and the volatility of social change. This tension underscores the imperative to recalibrate the alignment between evolving political values and China's modernization agenda. Achieving such balance is essential for sustaining social stability, reinforcing ideological legitimacy, and enabling rational and inclusive political value expression.

Key words: digital intelligence era; collective political values; individual political values; hierarchical structure; interactive socialization logic

Decision Interpretation Strategies and Public Trust in Algorithms——Survey Experiments for Administrative Approval Scenarios …… **Tan Haibo, Chu Qingqing, Wang Lin**(180)

Abstract: Artificial intelligence algorithms have been widely involved in decision-making tasks in various fields of the public sector, but the problem of "algorithmic trust crisis" in the public sector has not yet been solved. Based on the characteristics of AI algorithmic decision-making, this paper constructs a three-dimensional framework of "algorithm-organization-task" based on accessibility-diagnostic theory, signal theory and socio-technical system theory, and explores the impact of prior explanation of algorithms based on different types of information strategies on the public's perception of algorithmic trust in the public sector through a 2 (type of explanation strategy: one-sided information vs. two-sided information) × 2 (algorithmic decision-making method: augmented vs. automated) × 2 (decision complexity: high vs. low) scenario-based survey experiment (N = 1047) on the two types of administrative approvals, namely, the extension of the public health license and the recognition of teachers' qualifications. The results show that the information one-sidedness of algorithmic ex ante explanations had a significant effect on public algorithmic trust, and that algorithmic ex ante explanations based on one-sided information strategies were more conducive to enhancing public algorithmic trust than two-sided information strategies. Meanwhile, the algorithmic decision-making approach moderates the correlation between the type of explanation strategy and public algorithmic trust. In addition, the interaction effect of algorithmic decision interpretation strategy type and algorithmic decision-making approach on public perceived algorithmic trust is stronger as the decision complexity increases. Focusing on the characteristics of AI algorithmic decision-making, this paper

applies the accessibility-diagnosticity theory to explain the theoretical mechanism that one-sided information is more persuasive than two-sided information in the field of algorithmic decision-making and provides empirical evidence to help the public sector to choose appropriate explanatory strategies when deploying algorithmic decision-making.

Key words: algorithm explanation; double-sided information strategy; algorithm trust; scenario experiments

Theoretical Construction of International Political Communication in a New Era

…… **Jing Xuemin**(194)

Abstract: "International Political Communication" refers to the process of political information flow among countries, with the state serving as the political community (the main body of communication). The main contents of international political communication include ideology, political values, and political civilization. "International propaganda" represents the deep form of international political communication, "public diplomacy" serves as its media form, and "international political marketing" constitutes its extended form. The state-controlled mainstream media space is the main battlefield for international political communication, while the social media space, dominated by individuals, serves as a supplementary battlefield. The hybrid space, where mainstream and social media intersect, functions as a mixed battlefield, and platform-based media has emerged as a new battlefield. There are two models of international political communication: one is the horizontal, forceful penetration of political information, and the other is the vertical, downward flow of political information. Exploring the theoretical construction of "international political communication" in the new era holds significant implications for China. Chinese successful political practices challenge the traditional "cornerstones" of international political communication, correcting the problem of unbalanced traditional international political communication theory. The political civilization of socialism with Chinese characteristics embodied in the "Chinese path to modernization" provides a solid foundation for building new forms of international political communication in the new era. On this basis, it is essential to construct an independent discourse and knowledge system for international political communication that reflects Chinese distinctiveness, characteristics, and style.

Key words: new era; international political communication; Chinese political experience; Chinese path to modernization; political civilization

Urban Politics from the Global South Perspective: Conceptual Features, Theoretical Shifts, and Practical Innovations …… **Yu Zucheng, Peng Yang**(206)

Abstract: The development of Urban Politics from the perspective of the Global South reflects both a profound critique of traditional Western urban political frameworks and a response to the urgent need for localized research as Global South countries confront the challenges of rapid urbanization and social transformation. In this context, cities in the Global South, particularly those in China, have generated multi-layered innovations in urban political practices, encompassing global and cross-city cooperation, the restructuring of internal urban governance, and changes in modes of citizen participation. These innovations not only challenge the limitations of traditional Western urban political theories but also provide crucial empirical support for the theoretical development of Urban Politics in the Global South. However, research from this perspective still faces challenges such as an underdeveloped theoretical framework and insufficient explanatory power. Future research should integrate the theoretical innovations and practical experiences of cities in the Global South to construct a more systematic and analytically robust theoretical framework. This will promote the development of urban political science with Chinese characteristics, enhance its influence in global urban political studies, and provide solid theoretical support and policy guidance for urban governance in Global South countries.

Key words: global south; urban politics; theoretical shift; urban politics with Chinese characteristics

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决策解释策略与公众的算法信任*

——面向行政审批场景的调查实验研究

谭海波　褚清清　王　琳

内容提要：人工智能算法已广泛参与到公共部门各领域的决策任务中，但公众对算法决策的信任危机问题尚未解决。本文针对人工智能算法决策的特点，基于可及性－诊断性理论、信号理论和社会技术系统理论构建“算法－组织－任务”三维分析框架，通过公共卫生许可证延期和教师资格认定两种行政审批的2（解释策略类型：单面信息 vs. 双面信息）×2（算法决策方式：增强型 vs. 自动化型）×2（决策复杂度：高 vs. 低）情景调查实验（N＝1047），探讨了基于不同信息策略类型的算法事前解释对公众感知公共部门算法信任的影响。结果表明，算法事前解释的信息片面性对公众算法信任具有显著影响，基于单面信息策略的解释比双面信息策略更有利于增强公众的算法信任。同时，算法决策方式在解释策略类型与公众算法信任的关系中起调节作用。此外，随着决策复杂度提高，算法决策解释策略类型和算法决策方式对公众感知算法信任的交互效应更强。本文阐释了算法决策领域单面信息说服力度高于双面信息的理论机制并提供了经验证据，有助于公共部门部署算法决策时选择恰当的解释策略。

关键词：算法解释　双面信息策略　算法信任　情景实验

一、引言

信任是人际互动的基石，也是经济、社会和政治活动正常运行所必需的保障①。公众对公共部门的信任不仅包括对公职人员的信任，还包括对相关决策行为和制度机制的信任②。伴随着人工智能技术的日臻成熟和广泛应用，作为人工智能核心运作机制的算法开始广泛参与公共部门人员招聘、行政审批、灾害预警和行政执法等从日常到复杂的决策任务③，成为提升公共决策效率和质量的重要技术工具。传统街头官僚逐步转变为人机互动的屏幕官僚，进而演变为由算

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① 吴志成、李佳轩：《全球信任赤字治理的中国视角》，《政治学研究》，2020年第6期。

② Pippa Norris, *Critical Citizens: Global Support for Democratic Government*, New York: Oxford University Press, 1999, p. 10.

③ Madalina Busuioc, “Accountable Artificial Intelligence: Holding Algorithms to Account”, *Public Administration Review*, Vol. 81, No. 5, 2021.

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法和信息系统主导的系统官僚[①],对官僚合法性、行政自由裁量权和公共价值的实现产生了深远影响。与此同时,算法不透明性和自主性[②]等算法自身特性触发了大数据杀熟、算法歧视、行政正当程序失灵等诸多风险[③],并导致公众对算法决策的信任危机[④],引发了学术界和实务界的广泛关注。

人工智能算法是指能够模拟人类智能承担特定任务和功能的机器或计算机程序[⑤]。它凭借从经验、环境中学习和推理的能力,对外界的各种不同要求做出智慧化回应。当前,关于公共部门算法信任的研究主要沿着两条脉络展开,一部分研究从理论层面上探索塑造算法信任的治理模式与制度规制[⑥],另一部分研究则从实证角度检验公众是否信任算法决策以及算法信任的影响因素。已有研究发现,任务、组织和算法特征等不同层面的因素均可能对算法信任产生影响[⑦]。其中,算法透明度是影响公众算法信任的关键因素。在刑事司法[⑧]、签证申请、福利欺诈识别[⑨]、税务稽查[⑩]和企业政策性担保保费补贴申请[⑪]等情景中开展的实验研究均表明,算法透明度对于加强公众对算法决策的感知可信度至关重要。

提高算法透明度的重要途径之一是对算法进行解释,即用非技术语言向用户解释算法决策,以便用户能够理解、适当信任并有效管理人工智能结果[⑫]。关于算法解释与公众算法信任的研究大多从程序正义的角度出发,认为公众希望获得决策的明确理由,以评估决策的公平性,或能够对算法决策提出异议。因此,对算法进行解释可以提高公众的算法信任[⑬]。也有研究表明,公众可能会认为算法决策依赖于复杂的计算程序,即使提供了算法解释,他们也无法理解[⑭]。研究结论的异质性表明算法解释对公众算法信任的影响是复杂的。已有研究主要关注算法的事后解释,强调算法建议的可解释性。但最近的研究发现,使用算法的先前经验和对算法的熟悉程度同样重要。当公众了解数字技术时,他们在使用算法系统时可能会更有安全感和信心[⑮]。这意味

① 吴进进、何包钢:《算法科层制的兴起及其形态》,《社会学研究》,2023 年第 6 期。

② 张凌寒:《算法权力的兴起、异化及法律规制》,《法商研究》,2019 年第 4 期。

③ 亓光:《人工智能的社会公正风险:核心问题、理解视角与呈现类型》,《政治学研究》,2023 年第 6 期。

④ 董青岭:《人工智能时代的算法黑箱与信任重建》,《人民论坛·学术前沿》,2024 年第 16 期。

⑤ Bernd W. Wirtz, Jan C. Weyerer and Carolin Geyer, "Artificial Intelligence and the Public Sector—Applications and Challenges", *International Journal of Public Administration*, Vol. 42, No. 7, 2019.

⑥ 肖红军:《算法责任:理论证成、全景画像与治理范式》,《管理世界》,2022 年第 4 期。

⑦ 景怡:《算法态度文献综述、研究框架和未来展望》,《华东经济管理》,2023 年第 5 期。

⑧ Daniel S. Schiff, Kaylyn Jackson Schiff, Patrick Pierson, "Assessing Public Value Failure in Government Adoption of Artificial Intelligence", *Public Administration*, Vol. 100, No. 3, 2021.

⑨ Stephan Grimmelikhuijsen, "Explaining Why the Computer Says No: Algorithmic Transparency Affects the Perceived Trustworthiness of Automated Decision-Making", *Public Administration Review*, Vol. 83, No. 2, 2023.

⑩ 樊博、李晶晶:《算法透明何以提升公众的政府信任?》,《公共行政评论》,2024 年第 1 期。

⑪ 刘特、郑跃平、杨学敏:《公共部门中的算法透明能否影响公众信任? ——基于一项调查实验》,《电子政务》,2024 年第 8 期。

⑫ Pascal Hamm, Michael Klesel, Patricia Coberger, H. Felix Wittmann, "Explanation matters: An experimental study on explainable AI", *Electron Markets*, Vol. 33, 2023.

⑬ Stephan Grimmelikhuijsen, "Explaining Why the Computer Says No: Algorithmic Transparency Affects the Perceived Trustworthiness of Automated Decision-Making", *Public Administration Review*, Vol. 83, No. 2, 2023.

⑭ Kimberly D. Elsbach, Ileana Stigliani, "New Information Technology and Implicit Bias", *Academy of Management Perspectives*, Vol. 33, No. 2, 2019.

⑮ Manuel F. Gonzalez, Weiwei Liu, Lei Shirase, David L. Tomczak, Carmen E. Lobbe, Richard Justenhoven, Nicholas R. Martin, "Allying with AI? Reactions toward Human-Based, AI/ML-Based, and Augmented Hiring Processes", *Computers in Human Behavior*, Vol. 130, 2022.

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着对算法决策的相关信息提供事前解释可能会提高公众的算法信任。此外,决策解释所提供的信息量、措辞和特定词语的选择都可能影响个人的感知①。来自营销领域的研究表明,信息片面性(message sidedness)会影响信息接收主体的态度或信念②。但尚未有研究从公共部门的角度考虑单面信息与双面信息等具有不同信息片面性的决策解释策略对公众算法信任的影响。

总体而言,目前关于公众对公共部门算法信任的研究仍处于探索阶段,尤其是对算法决策事前解释类型与公众的算法信任之间的关系及其边界效应还有待进一步研究。基于此,本文围绕决策解释策略与公众算法信任的核心议题,整合可及性 - 诊断性理论、信号理论和社会技术系统理论构建"算法 - 组织 - 任务"三维分析框架,通过公共卫生许可证延期和教师资格认定两种行政审批的 2(解释策略类型:单面信息 vs. 双面信息)×2(算法决策方式:增强型算法决策 vs. 自动化算法决策)×2(决策复杂度:高 vs. 低)情景调查实验(N = 1047),检验信息片面性对公众算法信任的影响,以探索提高公众算法信任的决策解释策略。

二、理论基础与研究假设

(一)解释策略类型与公众的算法信任

只包含支持性论点的信息被称为单面信息,同时包含反对观点的信息被称为双面信息。单面信息或双面信息对态度或信念改变的说服效果被称为信息片面性效应③。关于信息片面性效应,来自不同领域的文献得出了不同结论。基于归因 - 对应理论(attribution-correspondence theory)的研究认为双面信息的说服力度优于单面信息。当信息具有双面性时,接收者倾向于接受正面主张更可能是由于产品属性和性能的有效性,而非别有用心地推销产品④。因此,当传播者承认自己的弱点时,接收者可能会更加信任信息⑤。尽管如此,在某些情况下,单面信息可能会被认为更可信,比如在党派新闻领域的研究发现,人们通常更喜欢态度一致的单面信息,部分原因是这类信息更容易启发式地处理⑥。由此可见,归因理论不足以解释其他领域尤其是公共领域的信息片面性效应。算法决策系统并非公共部门有意向公众推销的产品,而是已经实施的政策,因而公众不会向外归因,而是将信息中的优势主张向内归因于算法决策系统本身。

因此,鉴于算法的"黑箱"性质,本文采用可及性 - 诊断性理论(Accessibility-Diagnosticity Theory)对算法决策领域的信息片面性效应进行分析。该理论认为,用于决策的输入信息的价值取决于输入信息的可获取性和诊断性⑦。这两个维度对消费者行为的影响程度不同。当信息易

① Joy Lu, Dokyun Lee, Tae Wan Kim and David Danks, "Good Explanation for Algorithmic Transparency", *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 2020.

② Yousueng Han, "Message-Sidedness in Performance Information Disclosure and Citizens' Perceived Accountability: An Experimental Study", *Public Management Review*, 2024(online).

③ Marijke De Veirman, Liselot Hudders, "Disclosing Sponsored Instagram Posts: The Role of Material Connection with the Brand and Message-Sidedness When Disclosing Covert Advertising", *International Journal of Advertising*, Vol. 39, No. 1, 2019.

④ Robert E. Smith, Shelby D. Hunt, "Attributional Processes and Effects in Promotional Situations", *Journal of Consumer Research*, Vol. 5, No. 3, 1978.

⑤ Martin Eisend, "The Influence of Humor in Advertising: Explaining the Effects of Humor in Two-Sided Messages", *Psychology and Marketing*, Vol. 39, No. 5, 2022.

⑥ Axel Westerwick, Benjamin K. Johnson, Silvia Westerwick, "Confirmation Biases in Selective Exposure to Political Online Information: Source Bias vs. Content Bias", *Communication Monographs*, Vol. 84, No. 3, 2017.

⑦ Paul M. Herr, Frank R. Kardes, John Kim, "Effects of Word-of-Mouth and Product-Attribute Information on Persuasion: An Accessibility-Diagnosticity Perspective", *Journal of Consumer Research*, Vol. 17, No. 4, 1991.

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于或方便消费者检索时,可获取性就高,而当输入信息有助于消费者做出决策时,诊断性就高[①]。经验证据表明,可及性 - 诊断性理论有助于预测用于驱动行为结果的信息类型[②]。根据可及性 - 诊断性理论,在缺乏信息或无法获取相关信息的情况下,人们会根据直觉和预感行事。然而,当人们通过经验获得相关信息时,这些信息就会推翻他们无意识的偏见,更新他们有意识的信念,并增加对这些信念的依赖。

由于人工智能系统对于人类而言是一个不同的、充满不确定性的新"物种",其复杂的计算特点和特定的"黑箱"性质容易引发人们产生原始的无意识偏见[③]。当决策接受者接触到关于人工智能算法的性能信息时,他们会形成关于人工智能算法成功概率的主观估计,并在之后的相关决策任务中更多地依赖于学习到的期望概率而不是他们的内隐联想做出决策[④]。若决策接受者接触到的人工智能算法相关的性能信息是积极的单面信息,意味着他们感知到了更高的人工智能算法成功的期望概率,因而会减轻"算法厌恶"[⑤],增强对人工智能算法的信任。若决策接受者接触到的人工智能算法相关的性能信息是包含正面和负面观点的双面信息,由于负面信息更具诊断性,在形成总体评价时会得到更多的权重[⑥]。因此,相较于仅接受积极的单面信息的决策接受者,接受双面信息的决策接受者可能对人工智能算法成功的期望概率更低。由此,本文提出假设 1。

H1:与使用双面信息策略相比,使用单面信息策略对公共部门算法决策进行事前解释更有可能增加公众的算法信任。

(二)算法决策方式的调节作用

冈萨雷斯(Gonzalez)等通过判断人工智能算法决策系统是对人类决策者的补充还是取而代之,将人工智能算法决策系统与人类决策者的协作方式,也即算法决策方式,分为增强型算法决策和自动化算法决策两类[⑦]。信号理论认为,由于信息不对称,信号接收者(外部人员)将根据信号发送者(内部人员)发出的信号(内部信息)做出决策[⑧]。对于陌生产品,消费者无法判断其质量而难以产生信任,通过与市场上更受消费者信任的品牌或符号相联系,企业能够向消费者传递积极的信号,促进消费者信任[⑨]。本文在可及性 - 诊断性理论的基础上,融合信号理论分析不同

① Jyh-Shen Chiou, Cheng-Chieh Hsiao, Tien-Yi Chiu, "The Credibility and Attribution of Online Reviews: Differences Between High and Low Product Knowledge Consumers", *Online Information Review*, Vol. 42, No. 5, 2018.

② Greet Van Hoye, Filip Lievens, "Tapping the Grapevine: A Closer Look at Word-of-Mouth as a Recruitment Source", *The Journal of Applied Psychology*, Vol. 94, No. 2, 2009.

③ Ofir Turel, Shivam Kalhan, "Prejudiced Against the Machine? Implicit Associations and the Transience of Algorithm Aversion", *MIS Quarterly*, Vol. 47, No. 4, 2023.

④ Antoine Bechara, Antonio R. Damasio, "The Somatic Marker Hypothesis: A Neural Theory of Economic Decision", *Games and Economic Behavior*, Vol. 52, No. 2, 2005.

⑤ Antonya Marie Gonzalez, Jennifer R Steele, Evelyn F Chan, Sarah Ashley Lim, Andrew Scott Baron, "Developmental Differences in the Malleability of Implicit Racial Bias Following Exposure to Counter stereotypical Exemplars", *Developmental Psychology*, Vol. 57, No. 1, 2021.

⑥ Rohini Ahluwalia, Zeynep Gürhan-Canli, "The Effects of Extensions on the Family Brand Name: An Accessibility-Diagnosticity Perspective", *Journal of Consumer Research*, Vol. 27, No. 3, 2000.

⑦ Manuel F. Gonzalez, Weiwei Liu, Lei Shirase, David L. Tomczak, Carmen E. Lobbe, Richard Justenhoven, Nicholas R. Martin, "Allying with AI? Reactions toward Human-Based, AI/ML-Based, and Augmented Hiring Processes", *Computers in Human Behavior*, Vol. 130, 2022.

⑧ Brian L. Connelly, S. Trevis Certo, R. Duane Ireland, Christopher R. Reutzel, "Signaling Theory: A Review and Assessment", *Journal of Management*, Vol. 37, No. 1, 2011.

⑨ Adnan Muhammad Shah, Wazir Muhammad, KangYoon Lee, "Investigating the Effect of Service Feedback and Physician Popularity on Physician Demand in the Virtual Healthcare Environment", *Information Technology and People*, Vol. 36, No. 3, 2022.

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算法决策方式对解释策略类型与公众算法信任间关系的影响。

相较于人工智能,公众更信任人类决策者。在人工智能能力不确定的情况下,人类与人工智能的合作可被视为一种积极的信号。当不确定的人工智能算法决策系统与可靠的人类相结合时,它们可以向决策接受者传达各自无法独立提供的服务质量信息①。虽然决策接受者不确定人工智能算法决策系统能否有效回应和解决他们的请求,但他们相信与人工智能算法决策系统合作的人类可以做到,人类 - 人工智能团队合作的信号意味着人类员工可以改进人工智能相关工作②。因此,增强型算法决策能够增加决策接受者对算法决策的信任。在增强型算法决策的条件下,虽然决策接受者对双面信息的学习可能降低对人工智能算法成功的期望概率,但新增的"人类决策者可以为人工智能算法决策系统兜底"的可及信息减弱了负面信息的消极影响。此时,无论是学习了单面信息还是双面信息的决策接受者对人工智能算法决策系统的信任可能都较高。反之,在自动化算法决策条件下,不存在新增的与人类合作相关的可及信息,因而单面信息的说服效力要高于双面信息。由此,本文提出假设 2。

H2:算法决策方式在解释策略类型和公众算法信任的关系中起到调节作用。具体地,当算法决策方式为自动化型算法决策时,单面信息相较于双面信息对公众算法信任的影响更为显著。而当算法决策方式为增强型算法决策时,单面信息和双面信息对公众算法信任的影响差异较小。

(三)决策复杂度的高阶调节作用

根据社会技术系统(Sociotechnical Systems)理论,技术应用效果产生于技术、组织与情境的整合与协调③。技术属性与组织技术应用方式的协调取决于决策任务的情境约束,因此,具有不同复杂度的算法决策任务会影响算法解释策略与算法决策方式的组合效应。以往研究发现,决策复杂度决定了人们期望算法在决策中的参与深度。对于根据预测的客户量进行工作安排等简单的"定量"任务,算法和人类被认为是同样公平的④。算法可以提高准确性,而且使用最准确的方案也不会引起争议⑤。这意味着,对于决策复杂度较低的任务,公众并不在意人类是否参与算法决策。因此,算法决策方式采用自动化型还是增强型并不会改变公众接收到公共部门不同算法解释时的算法信任。

然而,随着复杂性的增加,公众会赞成在管理决策中采用更多的人为判断⑥。纳格特加尔(Nagtegaal)发现,对于复杂程度较低的实践,当决策由算法做出时,程序公正性会提高;对于复杂度较高的业务,与使用算法由计算机自动做出的决策相比,由公共管理者参与做出的决策被认为程序

① Jason W. Burton, Mari-Klara Stein, Tina Blegind Jensen, "A Systematic Review of Algorithm Aversion in Augmented Decision Making", *Journal of Behavioral Decision Making*, Vol. 33, No. 2, 2019.

② Ulrich Gnewuch, Stefan Morana, Oliver Hinz, Ralf Kellner, Alexander Maedche, "More Than a Bot? The Impact of Disclosing Human Involvement on Customer Interactions with Hybrid Service Agents", *Information Systems Research*, Vol. 35, No. 3, 2023.

③ T. A. Bentley, S. T. T. Teo, L. McLeod, F. Tan, R. Bosua, M. Gloet, "The Role of Organisational Support in Teleworker Wellbeing: A Socio-Technical Systems Approach". *Applied Ergonomics*, Vol. 52, 2016.

④ Min Kyung Lee, "Understanding Perception of Algorithmic Decisions: Fairness, Trust, and Emotion in Response to Algorithmic Management", *Big Data & Society*, Vol. 5, No. 1, 2018.

⑤ William M. Grove, Paul E. Meehl, "Comparative Efficiency of Informal (Subjective, Impressionistic) and Formal (Mechanical, Algorithmic) Prediction Procedures: The Clinical-Statistical Controversy", *Psychology, Public Policy and Law*, Vol. 2, No. 2, 1996.

⑥ Peter André Busch, Helle Zinner Henriksen, Øystein Sæbø, "Opportunities, Challenges of Digitized Discretionary Practices: A Public Service Worker Perspective", *Government Information Quarterly*, Vol. 35, No. 4, 2018.

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公正性更高[①],即:在低复杂性的决策场景下,公众更倾向于信任算法;而在高复杂性的决策场景下,公众倾向于赞成在管理决策中采用更多的人为判断。因此,对于决策复杂度较高的任务,人类是否参与算法决策对公众的算法信任至关重要,将对算法解释与算法信任的关系产生影响。对于决策复杂度较低的任务,算法被视为是以规则为基础的,因而能保持客观,并保证公众之间的平等待遇。在没有人类兜底的情况下,仅凭算法自身的优势就能够获取公众信任。而对于决策复杂度较高的任务,算法被认为无法替代人类判断,专业的自由裁量权更受青睐[②]。此时,仅强调算法优势这一可及信息无法改变公众对算法的隐形偏见,还需要增加人类参与的可及信息。由此,本文提出假设 3。

H3:解释策略类型、算法决策方式以及决策复杂度对公众算法信任存在三重交互作用。具体地,当决策复杂度较高时,解释策略类型和算法决策方式对公众感知算法信任存在显著的交互作用,在算法决策方式为自动化型算法决策的条件下,单面信息相较于双面信息对公众算法信任的影响比其他组合下的影响更大;而在算法决策方式为增强型算法决策的条件下,单面信息和双面信息对公众算法信任的影响差异较小。当决策复杂度较低时,解释策略类型和算法决策方式对公众算法信任的交互效应不显著。

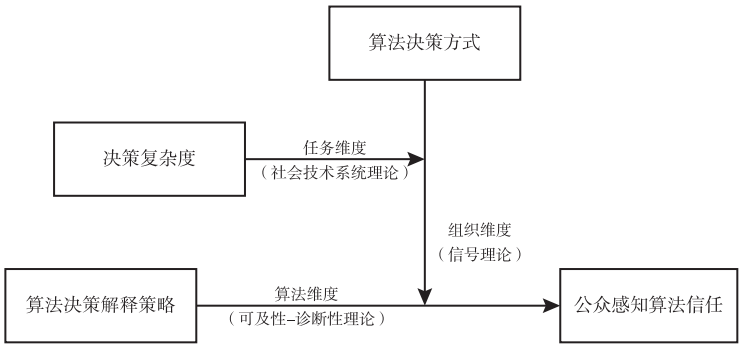


图 1　分析框架

三、研究设计

(一)研究对象与实验设计

本文采用情景实验方法进行假设检验。该方法有利于通过既定的情境控制实验条件,避免其他无关因素的干扰,减少"内生性"问题和偏误[③]。为增强实验的外部效度,作者使用已经在石家庄市、长沙市、青岛市等城市行政审批中应用的案例,并与电子政务专家、传播学专家、人工智能专家、政府官员以及行政审批事项申请者讨论情景设定的科学性。利用"见数(Credamo)"调研平台招募参与者。采用 G * power 3.1 软件[④]计算研究所需样本量,效应量设置为 0.25,α 设置

① Rosanna Nagtegaal, "The Impact of Using Algorithms for Managerial Decisions on Public Employees' Procedural Justice", *Government Information Quarterly*, Vol. 38, No. 1, 2021.

② Min Kyung Lee, "Understanding Perception of Algorithmic Decisions: Fairness, Trust, and Emotion in Response to Algorithmic Management", *Big Data & Society*, Vol. 5, No. 1, 2018.

③ 刘军强、胡国鹏、李振:《试点与实验:社会实验法及其对试点机制的启示》,《政治学研究》,2018 年第 4 期。

④ Franz Faul, Edgar Erdfelder, Albert-Georg Lang, Axel Buchner, "G * Power 3: A Flexible Statistical Power Analysis Program for the Social, Behavioral, and Biomedical Sciences", *Behavior Research Methods*, Vol. 39, No. 2, 2007.

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为 0.05,计算结果表明,为了达到 0.95 的统计检验力,研究至少需要 341 名被试。本研究实际招募了 1047 名被试。数据来源在地域上覆盖了广东、山东、江苏、河北、河南等 31 个省级行政区,研究样本具有广泛性和良好的代表性。在样本人口统计学特征分布方面,性别上,女性 702 人,男性 345 人;年龄上,平均年龄为 30.9 岁(SD = 8.6);学历上,大多数为本科学历,约占 77.7%,硕士及以上学历约占 18.2%;工作情况方面,大多数至少有一份稳定的正式工作,约占 77.3%;民族方面,被试者绝大多数为汉族,约占 93.0%。研究的被试样本在结构上呈现女性占比高、以有工作的青年为主等特点。

参与者被随机分入 2(解释策略类型:单面信息 vs. 双面信息)×2(算法决策方式:增强型 vs. 自动化型)×2(决策复杂度:高 vs. 低)组间设计的其中一种情境。问卷系统自动进行随机分配,作者并对样本进行了平衡性检验,以保证完全随机效果,避免其他变量对实验结果的影响。结果显示,性别、民族、年龄、受教育程度、就业状况等人口统计变量均不存在显著的组间差异,说明随机化实验得到了保证,样本平衡性检验通过。问卷首先向参与者说明本次调查的目的和注意事项,向参与者解释本研究中提到的"人工智能""行政审批""智能审批系统"等术语的含义,收集参与者接受实验操纵之前的初始算法态度和算法熟悉度等基本信息,紧接着告知参与者接下来将阅读一段材料和网页上的内容,并想象自己是材料中的"你"。材料分别对应算法解释策略类型、算法决策方式以及决策复杂度的相关情景。

(二)实验过程

在决策复杂度较低的情景中,参与者被告知自己正在经营一家理发店,正常经营所必需的公共场所卫生许可证即将到期,需要在当地政务服务网站上申请核验和延期。紧接着,参与者会看到参照政务服务网站智能审批模块网页设计构建的虚拟网页,分别根据实验条件设置不同的弹窗。其中,自动化算法决策与单面信息解释组的弹窗内容包括对智能审批系统的算法架构(如基于深度学习的 OCR 识别、图像对比和印章识别算法)、运行机制的解释以及"通过使用机器学习算法,该系统可以在两分钟内出具审核结果,大幅提高审批的效率和一致性,减少人为错误和主观偏见,同时提升审批的透明度和可追溯性"正向单面信息。双面信息组是在上述解释和单面信息的基础上,增加"但受技术限制和数据质量的影响,仍存在因材料识别失败而拒绝申请的可能"负向单面信息。增强型算法决策组是在上述解释和单/双面信息的基础上,增加"如果您对审批结果存在异议,可以继续提交人工审批"的人类参与信息。阅读材料后,参与者需要回答是否理解了实验场景、接受了实验处理和对算法决策的感知信任等问项。在决策复杂度较高的情景中,参与者被告知自己是一名即将毕业的本科生,刚刚通过教师资格考试,需要在当地政务服务网站上申请教师资格认定。其余流程与上述步骤一致,但弹窗内容中的申请材料改为"体检表""申请人所属人员范围材料""考试合格证明"等。

(三)变量测量

核心因变量是算法决策的感知可信度。目前关于决策接受者对人工智能信任的研究主要有两种测量信任的方法。第一种方法是根据人类信任度和技术信任度的三大支柱(仁慈、诚信和能力)构建的①。第二种方法是直接测量信任②,如表 1 题项所示。本文认为仁慈、诚信和能力只

① Hyesun Choung, Prabu David, Arun Ross, "Trust in AI and Its Role in the Acceptance of AI Technologies", *International Journal of Human-Computer Interaction*, Vol. 39, No. 9, 2022.

② Miriam Höddinghaus, Dominik Sondern and Guido Hertel, "The Automation of Leadership Functions: Would People Trust Decision Algorithms?", *Computers in Human Behavior*, Vol. 116, 2021.

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是影响算法信任的部分预测因素,并不能完全代表公众对算法的信任感知,因此采用第二种方法测量算法信任。该方法量表的 Cronbach's α 系数达到 0.92,KMO 指数为 0.85,$p < 0.05$,说明问卷具有良好的结构效度。需要注意的是,为了便于理解,问卷并没有从字面上提及“算法”一词,而是“智能计算机系统”。

已有研究认为决策接受者的性别①、种族②、被歧视经历③、对结果的控制程度④、对算法的熟悉程度⑤等因素可能影响其对算法的态度。本文参照上述研究设置控制变量(见表 2)。其中,对于算法熟悉度,本文借鉴特奥多雷斯库(Teodorescu)等⑥的研究,通过询问受访者是否具有算法相关的专业知识进行测量。此外,本文还控制了被试原有的算法使用意愿。

表 1　感知可信度量表

题项	测量
我非常依赖于智能计算机系统。 我会完全信任智能计算机系统。 我依赖于智能计算机系统会让我感到舒服。	采用 likert 七级打分法,1 表示“非常不赞同”,7 表示“非常赞同”。

表 2　控制变量与测量

变量	测量
性别	男 =1,女 =0
民族	汉族 =1,其他 =0
年龄	34 岁及以下 =0;35—59 岁 =1;60 岁及以上 =2
受教育程度	高中及以下 =0;大学 =1;研究生及以上 =2
就业状况	有工作 =1;没有工作 =0
智能审批服务使用经历	是 =1,否 =0
算法专业知识	是 =1,否 =0
算法使用意愿	非常愿意 =5,比较愿意 =4,一般 =3,比较不愿意 =2,非常不愿意 =1

① Florian Pethig, Julia Kroenung, “Biased Humans, (Un) Biased Algorithms?”, *Journal of Business Ethics*, Vol. 183, No. 3, 2022.

② Rewina Bedemariam, Jennifer L. Wessel, “The Roles of Outcome and Race on Applicant Reactions to AI Systems”, *Computers in Human Behavior*, Vol. 148, 2023.

③ Irmela F. Koch-Bayram, Chris Kaibel, Torsten Biemann, María del Carmen Triana, “</Click to Begin Your Digital Interview>: Applicants' Experiences with Discrimination Explain Their Reactions to Algorithms in Personnel Selection”, *International Journal of Selection and Assessment*, Vol. 31, No. 2, 2023.

④ David T. Newman, Nathanael J. Fast, Derek J. Harmon, “When Eliminating Bias Isn't Fair: Algorithmic Reductionism and Procedural Justice in Human Resource Decisions”, *Organizational Behavior and Human Decision Processes*, Vol. 160, 2020.

⑤ Manuel F. Gonzalez, Weiwei Liu, Lei Shirase, David L. Tomczak, Carmen E. Lobbe, Richard Justenhoven, Nicholas R. Martin, “Allying with AI? Reactions toward Human-Based, AI/ML-Based, and Augmented Hiring Processes”, *Computers in Human Behavior*, Vol. 130, 2022.

⑥ Daniel Teodorescu, Kamer-Ainur Aivaz, Diane Paula Corine Vancea, Elena Condrea, Cristian Dragan, Ana Cornelia Olteanu, “Consumer Trust in AI Algorithms Used in E-Commerce: A Case Study of College Students at a Romanian Public University”, *Sustainability*, Vol. 15, No. 15, 2023.

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四、结果与分析

(一)操纵检验

本文采用直接操纵检验(direct manipulation)和指导式操纵检验(instructional manipulation checks)两种方式检验实验操纵有效性。首先,检验信息策略操作的有效性。与之前的研究一致,我们通过以下项目来测量信息的双面性——“这条信息不仅提出了智能计算机系统可以提供良好服务的论点,还考虑了它们仍存在一些不足的论点”①,采用七点语义差别量表。结果显示,参与者对双面信息(M = 5.97,SD = 0.78)的双面性评价高于单面信息(M = 4.93,SD = 1.27,$p < 0.01$,Cohen's d = 0.98)。因此,在正式实验中,对信息策略的操纵是成功的,也是有用的。其次,根据豪泽(Hauser)等②的研究,采用注意力检测的方式进行指导式操纵检验。在量表中插入注意力检测题项“本题检测是否认真作答,请选 1”,只保留在该题项中做出正确选择的样本数据(剔除率 4.2%),以确保被试集中注意力,认真填答。

(二)假设检验

为检验假设 1,对算法决策解释策略类型(0 = 单面信息策略,1 = 双面信息策略)和公众感知算法信任进行回归分析。结果显示,采用双面信息解释策略的算法决策,其公众感知算法信任显著低于单面信息解释策略组(b = -0.48,$p < 0.01$)(见表 3 模型 2),公共部门应用算法决策时采用单面信息解释策略比采用双面信息解释策略更能够提高公众对公共部门算法决策的信任。因此,假设 1 得证。

表 3 算法决策解释策略类型对公众算法信任的影响

变量	算法信任	
	模型 1	模型 2
算法决策解释策略类型	-0.56 *** (0.08)	-0.48 *** (0.07)
控制变量		控制
常数项	5.08 *** (0.05)	2.03 *** (0.25)
样本量	1047	1047
adj. R^2	0.05	0.24

注:* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$。

为检验假设 2,分别做算法决策解释策略类型、算法决策方式(0 = 自动化型算法决策,1 = 增强型算法决策)及其交互项对公众感知算法信任的回归分析。如表 4 模型 2 所示,解释策略类型与算法决策方式的交互项系数显著(b = 0.23,$p < 0.1$),表明解释策略类型与算法决策方式对公

① Erlinde Cornelis, Veroline Cauberghe, Patrick De Pelesmacker, “Regulatory Congruence Effects in Two-Sided Advertising: The Mediating Role of Processing Fluency and Processing Depth”, *European Journal of Marketing*, Vol. 48, No. 7, 2014.

② David J Hauser, Phoebe C Ellsworth, Richard Gonzalez, “Are Manipulation Checks Necessary?”, *Frontiers in Psychology*, Vol. 9, 2018.

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众感知算法信任存在交互作用。进一步地,简单效应分析(见图 2)发现,当算法决策方式为自动化型时,采用双面信息解释策略的算法决策,其公众感知算法信任显著低于单面信息解释策略组($b = -0.51, p < 0.01$);当算法决策方式为增强型时,采用双面信息解释策略的算法决策,其公众感知算法信任仍显著低于单面信息解释策略组($b = -0.28, p < 0.1$),但效应量较小。因此,假设 2 得证。

表 4　算法决策解释策略类型与算法决策方式交互项对公众算法信任的影响

变量	算法信任	
	模型 1	模型 2
算法决策解释策略类型	-0.51*** (0.07)	-0.63*** (0.10)
算法决策方式	0.24*** (0.07)	0.14 (0.09)
算法决策解释策略类型 × 算法决策方式		0.23* (0.14)
控制变量	控制	控制
常数项	1.95*** (0.25)	2.01*** (0.26)
样本量	1047	1047
adj. R^2	0.25	0.25

注:* $p < 0.1$;** $p < 0.05$;*** $p < 0.01$。

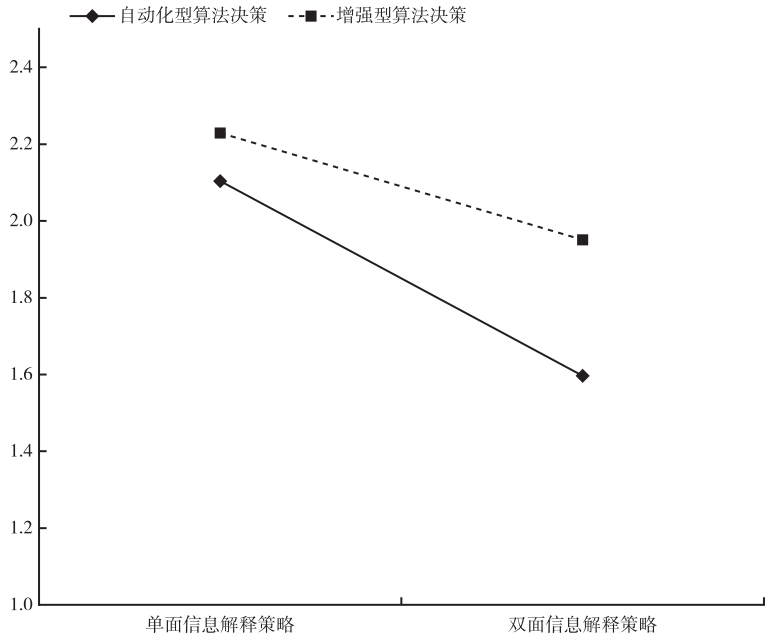


图 2　算法决策解释策略类型与算法决策方式对公众算法信任的交互作用

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为检验假设 3,分别做算法决策解释策略类型、算法决策方式、决策复杂度(0 = 低决策复杂度,1 = 高决策复杂度),两两之间的交互项,及其三重交互项对公众感知算法信任的回归分析。表 5 模型 2 的结果显示,算法决策解释策略类型、算法决策方式及决策复杂度的三重交互项系数显著(b = 0.74,p < 0.01)。进一步地,简单效应分析(见图 3)发现,当决策复杂度较高时,解释策略类型和算法决策方式对公众感知算法信任存在显著的交互作用(b = 0.72,p < 0.01)(见表 7 模型 1)。具体地,当算法决策方式为自动化型时,采用双面信息解释策略的算法决策,其公众感知算法信任显著低于单面信息解释策略组(b = -1.18,p < 0.01),即单面信息相较于双面信息对公众算法信任的影响比其他组合下的影响更大;当算法决策方式为增强型时,采用双面信息解释策略的算法决策,其公众感知算法信任仍显著低于单面信息解释策略组,但效应量较小(b = -0.46,p < 0.01),即单面信息和双面信息对公众算法信任的影响差异较小。当决策复杂度较高时,不同算法决策方式下算法决策解释策略对算法信任的影响差异显著(b = 0.72,p < 0.01)(见表 6)。相反地,当决策复杂度较低时,算法决策解释策略类型和算法决策方式对公众感知算法信任的交互效应并不显著(b = 0.02,p > 0.1)(见表 7)。因此,假设 3 得证。

表 5　算法决策解释策略类型、算法决策方式、决策复杂度对公众算法信任的影响

变量	算法信任	
	模型 1	模型 2
算法决策解释策略类型	-0.53*** (0.07)	-0.36*** (0.14)
算法决策方式	0.25*** (0.07)	0.20 (0.14)
决策复杂度	-0.14* (0.07)	0.07 (0.13)
算法决策解释策略类型 × 算法决策方式		-0.02 (0.19)
算法决策解释策略类型 × 决策复杂度		-0.82*** (0.21)
算法决策方式 × 决策复杂度		-0.12 (0.18)
算法决策解释策略类型 × 算法决策方式 × 决策复杂度		0.74*** (0.28)
控制变量	控制	控制
常数项	2.10*** (0.26)	2.15*** (0.27)
样本量	1047	1047
adj. R^2	0.25	0.27

注:* p < 0.1;** p < 0.05;*** p < 0.01。

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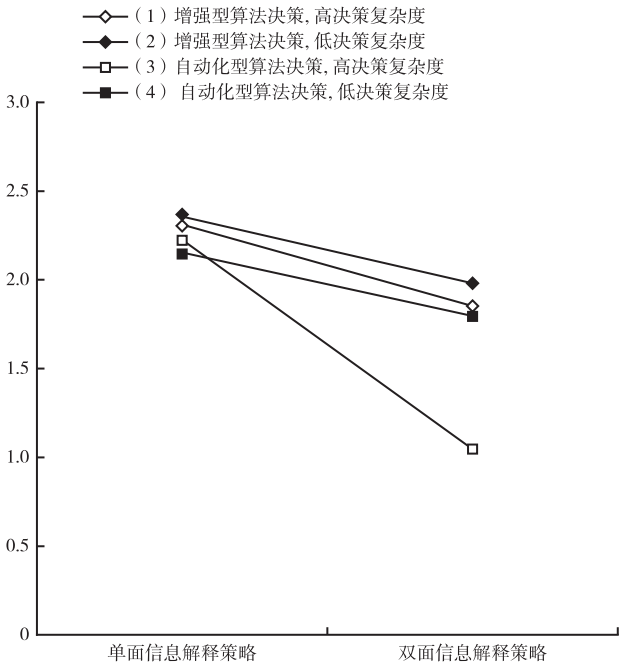


图 3　算法决策解释策略类型、算法决策方式和决策复杂度对公众算法信任的三重交互作用

表 6　四种情况下的回归斜率差异

组合	差值	95% 置信区间
条件 1 – 条件 3	0. 72	[0. 302，1. 137]
条件 1 – 条件 2	– 0. 083	[– 0. 450，0. 284]
条件 1 – 条件 4	– 0. 101	[– 0. 471，0. 270]
条件 2 – 条件 3	0. 803	[0. 385，1. 220]
条件 2 – 条件 4	– 0. 018	[– 0. 386，0. 350]
条件 3 – 条件 4	– 0. 821	[– 1. 240， – 0. 402]

表 7　不同决策复杂度组中算法决策解释策略类型、算法决策方式对公众算法信任的交互作用

变量	算法信任	
	高决策复杂度组	低决策复杂度组
算法决策解释策略类型	– 1. 21 ***	– 0. 38 ***
	（0. 17）	（0. 09）
算法决策方式	0. 10	0. 17
	（0. 13）	（0. 13）
算法决策解释策略类型 × 算法决策方式	0. 72 ***	0. 02
	（0. 22）	（0. 19）
控制变量	控制	控制
常数项	2. 49 ***	1. 76 ***
	（0. 36）	（0. 39）
样本量	491	556
adj. R^2	0. 28	0. 24

注：* $p < 0.1$；** $p < 0.05$；*** $p < 0.01$。

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五、结论与启示

(一)研究结论

随着人工智能技术的迅猛发展和广泛应用,算法正逐步渗入个人感知、商业交易甚至公共决策之中,成为公共部门的重要决策支持主体。而公众作为算法决策的接受者普遍存在"算法厌恶",对算法不透明等特性及其相关风险的忧虑消解了公众对算法决策的信任。对算法进行解释是提高算法透明度的重要途径。近年来,越来越多的学者深入微观与实践层面研究不同类型算法解释的信任效应。但事前算法决策解释策略与公众算法信任之间的关系及其边界效应还有待深入研究。

本文以公共卫生许可证延期和教师资格认定两种行政审批为实验场景,通过 2(解释策略类型:单面信息 vs. 双面信息)×2(算法决策方式:增强型 vs. 自动化型)×2(决策复杂度:高 vs. 低)情景调查实验(N = 1047)探讨了基于不同信息策略类型的算法事前解释对公众感知公共部门算法信任的影响。研究发现,与双面信息策略在其他领域的效力不同,在公共部门算法决策领域,基于单面信息策略的算法事前解释比双面信息策略更有利于增强公众对算法的信任。同时,算法决策方式在解释策略类型与公众算法信任的关系中起调节作用,相较于增强型算法决策,当算法决策方式为自动化型算法决策时,单面信息解释策略对公众算法信任的提升效应更强。此外,随着决策复杂度提高,解释策略类型和算法决策方式对公众感知算法信任的交互效应更强。

(二)理论贡献

第一,本研究揭示了事前算法解释策略的信任效应。以往关于算法解释的研究主要关注算法的事后解释,而为算法决策提供事前解释对公众算法信任的影响是未知的。本研究发现,策略性地提供算法事前解释能够增加公众对算法决策的信任。该结论证明了事前解释的重要意义,强调算法解释研究不能局限于事后解释,在决策发生前对算法系统的决策机制和优势进行解释亦能够增强算法信任,为后续开展算法事前解释的深入研究提供了基础。

第二,本研究拓展了信息片面性在算法决策领域的应用。已有研究初步探讨了程序解释、理由解释等不同算法决策解释类型的信任效应[①],但对于增加公众信任的有效解释类型仍存在争议。本研究创新性地从信息片面性的角度对算法解释进行分类,同时,考虑到公共部门算法决策的政策特性,基于可及性 - 诊断性理论提出了算法决策领域单面信息说服力度高于双面信息的新假设,并在公共卫生许可证延期和教师资格认定两种行政审批实验场景下完成了验证。这就初步证实了可及性 - 诊断性理论在信息片面性研究中的适用性,给出了人工智能算法决策在特定场域下与以往研究不同的新结论和新解释,拓展了关于信息片面性效应适用条件的理论认知。

第三,本研究证明了算法决策方式和决策复杂度在解释策略类型与公众算法信任关系中的复杂交互作用。以往关于算法信任的研究分别讨论了任务特征[②]、组织因素[③]和算法特征[④]等不

① E. N. Nieuwenhuizen, A. J. Meijer, F. J. Bex, S. G. Grimmelikhuijsen, "Explanations Increase Citizen Trust in Police Algorithmic Recommender Systems: Findings from Two Experimental Tests", *Public Performance & Management Review*, Vol. 48, No. 3, 2025.

② Rosanna Nagtegaal, "The Impact of Using Algorithms for Managerial Decisions on Public Employees' Procedural Justice", *Government Information Quarterly*, Vol. 38, No. 1, 2021.

③ Manuel F. Gonzalez, Weiwei Liu, Lei Shirase, David L. Tomczak, Carmen E. Lobbe, Richard Justenhoven and Nicholas R. Martin, "Allying with AI? Reactions toward Human-Based, AI/ML-Based, and Augmented Hiring Processes", *Computers in Human Behavior*, Vol. 130, 2022.

④ 樊博、李晶晶:《算法透明何以提升公众的政府信任?》,《公共行政评论》,2024 年第 1 期。

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同层面的因素对算法信任的影响,少数研究考虑了组织因素与算法特征的交互影响,但尚未有研究考虑不同层面影响因素之间的高阶交互作用。本研究基于可及性-诊断性理论、信号理论和社会技术系统理论构建了“算法-组织-任务”三维分析框架,证明了不同层面影响因素之间的复杂性关系,启示算法决策研究应当从整体视角充分考虑任务因素、组织因素与技术因素的相互作用,弥补了已有研究缺乏考虑复杂情景因素的不足。

(三)实践启示

第一,公共部门应加强算法透明的制度供给和策略设计。公共部门在设计算法解释的政策方案时应强调算法决策在效率、准确性、客观性等方面的传统决策方式无法比拟的优势,帮助公众理解算法决策的价值,从而减轻公众对算法参与公共决策的担忧。更重要的是,政府应注重持续提升公共部门算法设计的科学性和有效性,减少决策错误、算法偏见和隐私风险等问题,以便在设计算法解释的政策方案时突出算法优势,减少公众对算法“黑箱”的疑虑,增强公众对算法决策的理解与信任。

第二,公共部门应在不同的信息公开时点选择不同的解释策略。本研究聚焦算法的事前解释,即公众使用系统之前的阶段,结论表明事前解释应当向公众展示算法的性能及其优势以增强公众的信任,促进其使用智能审批系统。而在公众使用系统之后,即进入事后解释的阶段,若公众对系统的决策结果存在异议,公共部门应当启动事后解释的程序,解释算法做出决策的理由和依据,包括算法可能存在的风险等。因此,公共部门算法解释策略的选择并非随心所欲,或者一成不变,而是应当在合适的时机采用恰当的解释策略。

第三,公共部门需注重协调算法决策者与人类决策者的合作关系。提高算法信任度和接受度的方法之一是将算法作为与人类伙伴合作的产物,或者至少在人类的监督之下①。本文的研究结论亦表明,相较于算法独立做出决策的自动化型算法决策,在以人类为主导的增强型算法决策条件下,人们对算法的信任程度更高。尤其是在复杂度较高的决策情景下,公众倾向于赞成在管理决策中采用更多的人为判断。因此,公共部门应在强调算法决策优势的同时注重发挥人类决策者对决策质量的保障作用。

(四)局限与展望

本文仍存在一些局限,有待于未来进一步研究。第一,本文选取公共卫生许可证延期和教师资格认定两个案例作为实验场景,而未来研究可进一步考虑不同领域、不同类型算法的决策应用,设计、检验并比较相应的算法解释策略,探索信任的生成机制和路径。第二,本文在设计算法解释信息的双面性时,仅考虑了算法功能方面的特征,尚未将算法的隐私风险、情感特征等可能存在的负面信息纳入考量,未来可拓展算法双面信息类型,探索不同类型的双面算法解释策略的信任效应。第三,本文的实验情景主要聚焦面向公众的政务服务决策,未来可以对政治场景中涉及多元主体、竞争性利益、复杂博弈过程的宏观决策开展进一步的研究。

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① David De Cremer, Jack McGuire, “Human-Algorithm Collaboration Works Best If Humans Lead (Because It Is Fair!)”, *Social Justice Research*, Vol. 35, No. 1, 2022.

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amid change." On the one hand, digital technologies reinforce collective political values by strengthening national identity; on the other, they foster pluralistic expressions at the individual level. Yet, the accelerating pace of socio-economic development introduces frictions between the inertia of political value formation and the volatility of social change. This tension underscores the imperative to recalibrate the alignment between evolving political values and China's modernization agenda. Achieving such balance is essential for sustaining social stability, reinforcing ideological legitimacy, and enabling rational and inclusive political value expression.

Key words: digital intelligence era; collective political values; individual political values; hierarchical structure; interactive socialization logic

Decision Interpretation Strategies and Public Trust in Algorithms——Survey Experiments for Administrative Approval Scenarios …… **Tan Haibo, Chu Qingqing, Wang Lin**(180)

Abstract: Artificial intelligence algorithms have been widely involved in decision-making tasks in various fields of the public sector, but the problem of "algorithmic trust crisis" in the public sector has not yet been solved. Based on the characteristics of AI algorithmic decision-making, this paper constructs a three-dimensional framework of "algorithm-organization-task" based on accessibility-diagnostic theory, signal theory and socio-technical system theory, and explores the impact of prior explanation of algorithms based on different types of information strategies on the public's perception of algorithmic trust in the public sector through a 2 (type of explanation strategy: one-sided information vs. two-sided information) × 2 (algorithmic decision-making method: augmented vs. automated) × 2 (decision complexity: high vs. low) scenario-based survey experiment (N = 1047) on the two types of administrative approvals, namely, the extension of the public health license and the recognition of teachers' qualifications. The results show that the information one-sidedness of algorithmic ex ante explanations had a significant effect on public algorithmic trust, and that algorithmic ex ante explanations based on one-sided information strategies were more conducive to enhancing public algorithmic trust than two-sided information strategies. Meanwhile, the algorithmic decision-making approach moderates the correlation between the type of explanation strategy and public algorithmic trust. In addition, the interaction effect of algorithmic decision interpretation strategy type and algorithmic decision-making approach on public perceived algorithmic trust is stronger as the decision complexity increases. Focusing on the characteristics of AI algorithmic decision-making, this paper

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applies the accessibility-diagnosticity theory to explain the theoretical mechanism that one-sided information is more persuasive than two-sided information in the field of algorithmic decision-making and provides empirical evidence to help the public sector to choose appropriate explanatory strategies when deploying algorithmic decision-making.

Key words: algorithm explanation; double-sided information strategy; algorithm trust; scenario experiments

Theoretical Construction of International Political Communication in a New Era

…… **Jing Xuemin**(194)

Abstract: "International Political Communication" refers to the process of political information flow among countries, with the state serving as the political community (the main body of communication). The main contents of international political communication include ideology, political values, and political civilization. "International propaganda" represents the deep form of international political communication, "public diplomacy" serves as its media form, and "international political marketing" constitutes its extended form. The state-controlled mainstream media space is the main battlefield for international political communication, while the social media space, dominated by individuals, serves as a supplementary battlefield. The hybrid space, where mainstream and social media intersect, functions as a mixed battlefield, and platform-based media has emerged as a new battlefield. There are two models of international political communication: one is the horizontal, forceful penetration of political information, and the other is the vertical, downward flow of political information. Exploring the theoretical construction of "international political communication" in the new era holds significant implications for China. Chinese successful political practices challenge the traditional "cornerstones" of international political communication, correcting the problem of unbalanced traditional international political communication theory. The political civilization of socialism with Chinese characteristics embodied in the "Chinese path to modernization" provides a solid foundation for building new forms of international political communication in the new era. On this basis, it is essential to construct an independent discourse and knowledge system for international political communication that reflects Chinese distinctiveness, characteristics, and style.

Key words: new era; international political communication; Chinese political experience; Chinese path to modernization; political civilization

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